School Management System

Technology Stack & Deployment Decision

Final, locked decision: Next.js on Vercel + Spring Boot and PostgreSQL on Railway

October 2026

Status: **FINAL - DECISION LOCKED** (October 2026)

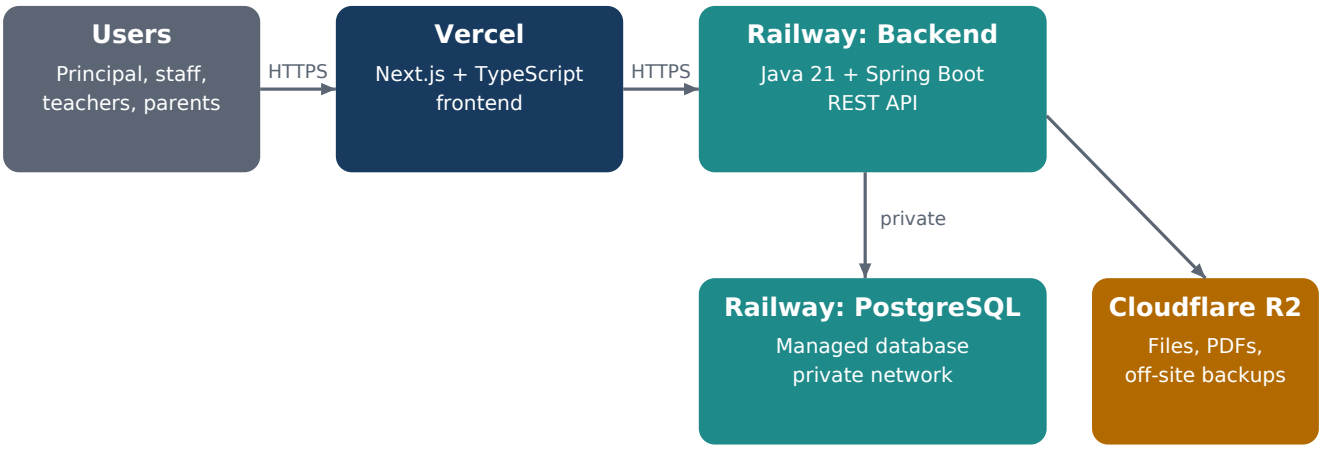
Prepared for: SelectCode - development of the school management system described in the Executive Functional Requirements.

Decision in one line: Next.js + TypeScript frontend on **Vercel**; **Java 21 + Spring Boot** backend and **PostgreSQL** database on **Railway (paid plan)**; **Cloudflare R2** for files and off-site backups.

1. Final Decision Summary

Layer	Locked choice	Hosted on
Frontend	**Next.js + React + TypeScript**, shadcn/ui or Mantine components	**Vercel**
Backend	**Java 21 + Spring Boot** (Spring Security, Spring Batch, Spring Data JPA, Flyway)	**Railway** (paid plan)
Database	**PostgreSQL**	**Railway** (same project and region as the backend)
Files and backups	Student photos, documents, generated PDFs, daily database dumps	**Cloudflare R2** (or Amazon S3)
Reports and exports	JasperReports or OpenPDF for PDFs; Apache POI for Excel	Runs inside the backend
Urdu support	Noto Nastaliq Urdu font, right-to-left layout	Frontend and PDF templates

Architecture



Users open the Next.js app on Vercel. The app calls the Spring Boot API on Railway over HTTPS. The API talks to PostgreSQL over Railway's private network, so the database is not exposed to the internet. Files and backups are stored in Cloudflare R2.

2. Why This Stack

The system is an ERP-style, money-heavy application with about 150 features across 16 modules. The choice is based on the needs of the system, not on any one developer's existing skills.

Requirement	How the stack meets it
Exact money	Java **BigDecimal** and PostgreSQL **numeric** types store fees and balances exactly.
Payments saved completely or not at all	Spring transactions save the receipt, balance and account entry together or not at all.
Roles and permissions	Spring Security: roles, per-class access, method-level rules, login history.
Monthly invoice runs and payroll	Spring Batch and scheduled tasks with restart and error handling.
Vouchers, report cards, Excel	JasperReports or OpenPDF for PDFs; Apache POI for Excel.
Long-term maintenance and handover	Strong typing, stable ecosystem, widely available Java developers.
Accessible, fast frontend	Next.js with React and accessible component libraries; WCAG practices.

Alternatives considered

Option	Verdict	Reason
C# / .NET 8	Equal alternative	On par with Java technically; not chosen.
Node.js / TypeScript	Second	Fast to build; weaker for exact money and mature reporting.
PHP (Laravel), Python (Django)	Workable	Weaker type safety for a system of this size.
Go, C++, Rust	Not suitable	Lack business-reporting tools or add complexity the project does not need.

3. Why Vercel + Railway

- **Simple deployment:** both deploy straight from GitHub; no server to maintain.
- **Backend and database together:** Spring Boot reaches PostgreSQL over Railway's private network - fast and not exposed to the internet.
- **No sleeping:** the paid Railway plan keeps the backend running, so there are no slow first requests in the morning.
- **Scales later:** CPU and memory can be increased, or more backend copies added, without redesign.
- **Reasonable cost** for one or a few schools.

4. Required Setup Rules

Area	Rule
Region	Backend and database in the same Railway region, the one closest to Pakistan (Railway lists a Singapore region - confirm availability on the plan).
Database backups	Enable Railway Postgres backups AND an automatic daily database dump to Cloudflare R2 or Amazon S3. Test a restore regularly.
Java memory	Set the Java memory limit to fit the Railway plan so the service does not use (and bill) more memory than needed.
Domains	Custom domains for both, e.g. **app.yourschool.com** (Vercel) and **api.yourschool.com** (Railway).
Security	HTTPS everywhere; the backend accepts requests only from the Vercel domain (CORS); secure, HTTP-only login cookies; secrets kept in environment variables, never in code.
Files	Photos, documents and generated PDFs stored in Cloudflare R2, not on the Railway server's disk.
Environments	Separate **staging** (testing) and **production** (live) environments on both Vercel and Railway.
Costs	Railway bills by usage: set a spending limit and review usage monthly.
Monitoring	Health checks, error alerts and logs for the backend; uptime monitoring for the API and website.

5. Deployment Workflow

Step	What happens
1. Code	Developer pushes changes to GitHub.
2. Staging	Vercel and Railway deploy automatically to staging; database changes applied by Flyway.
3. Test	The team checks the change on staging with test data.
4. Release	Changes are merged to the main branch and deployed to production.
5. Backup and monitor	Daily backups to R2; alerts on errors; rollback to the previous version if needed.

6. When to Reconsider

- If many schools are added and Railway's monthly bill becomes high, a dedicated server or AWS/Azure may become cheaper.
- If a school requires its data to stay in a specific country, check Railway's regions before onboarding it.
- If Railway's pricing or terms change significantly, the same Docker-based backend can move to another host with little change.

Decision locked: Next.js + TypeScript on Vercel; Java 21 + Spring Boot and PostgreSQL on Railway (paid plan); Cloudflare R2 for files and off-site backups.

Sources

Railway regions - alchemy.run/railway/compute/regions
Railway Postgres backups and restores - docs.railway.com/guides/postgres-backups-restores
Railway Postgres daily backups template - railway.com/deploy/postgres-daily-backups